

Barrier Option Pricing with Variance Reduction via Importance Sampling

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Outline

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What Is a Barrier Option?

- A **path-dependent** derivative: payoff depends on the full price history
- Two classes:
 - **Knock-in**: option *activates* if asset hits a barrier level
 - **Knock-out**: option *expires* if asset hits a barrier level
- We focus on an **up-and-in put option**:
 - Expires worthless if $S_t < B_u$ throughout
 - Becomes a vanilla put if $S_t \geq B_u$ at any monitoring date

Key Parameters

$S_0 = 100$	Initial price
$K = 100$	Strike
$B_u = 120$	Upper barrier
$r = 0.05$	Risk-free rate
$\sigma = 0.2$	Volatility
$T = 1$	Maturity
$n = 252$	Monitoring steps

Black-Scholes Model & Analytical Price

Underlying dynamics (risk-neutral measure):

$$\frac{dS_t}{S_t} = r dt + \sigma dW_t$$

Closed-form price (Rubinstein & Reiner, 1991):

$$p_i(S, t) = -S \left(\frac{B_u}{S} \right)^{2\mu+1} \mathcal{N}(-y_1) + Ke^{-rT} \left(\frac{B_u}{S} \right)^{2\mu} \mathcal{N}(-y_1 + \sigma\sqrt{T})$$

$$\mu = \frac{r - \frac{1}{2}\sigma^2}{\sigma^2}, \quad y_1 = \frac{\ln\left(\frac{B_u^2}{SK}\right)}{\sigma\sqrt{T}} + (1 + \mu)\sigma\sqrt{T}$$

Limitation

This formula assumes **continuous** barrier monitoring. Real contracts typically specify **discrete** monitoring instants $\Rightarrow$ price differences can be substantial.

Broadie–Glasserman–Kou Correction

Idea: Keep the continuous formula but *shift* the barrier to correct for discrete monitoring:

$$V_m(B_u) = V\left(B_u e^{\beta\sigma\sqrt{T/m}}\right) + o\left(\frac{1}{\sqrt{m}}\right)$$

where $\beta \approx 0.5826$ and m is the number of monitoring instants.

Result for our parameters:

- Analytical price: **0.2134**
- Corrected price: **0.1774**

Why it matters

Continuous vs. discrete can produce **substantial price differences**. The correction brings the analytical benchmark in line with simulation.

Importance Sampling — Core Idea

Problem: An up-and-in put pays off only when a rare joint event occurs:

- ① Asset *hits* the upper barrier B_u
- ② Asset *ends below* strike K at maturity

⇒ Standard Monte Carlo wastes most paths on zero-payoff scenarios.

Solution — two-phase drift shift:

Phase 1 — Pre knock-in

Shift drift *upward* to increase the probability of hitting B_u :

$$\theta_+ = \frac{1}{2} - \frac{r}{\sigma^2} + \frac{2b + c}{T\sigma^2}$$

$$b = \ln(B_u/S_0), \quad c = \ln(S_0/K)$$

Phase 2 — Post knock-in

Shift drift *downward* to push S_T below K :

$$\theta_- = \frac{1}{2} - \frac{r}{\sigma^2} - \frac{2b + c}{T\sigma^2}$$

Likelihood Ratio & IS Estimator

Girsanov correction — log-likelihood ratio for each path:

$$\log \mathcal{L} = (\theta_- - \theta_+) X_\tau - \theta_- X_T + n \psi_-, \quad \psi_- = \mu \theta_- + \frac{1}{2} \sigma^2 \Delta t \theta_-^2$$

IS price estimator:

$$\hat{V}^{\text{IS}} = \frac{1}{m} \sum_{i=1}^m e^{-rT} \max(K - S_T^{(i)}, 0) \mathbf{1}_{\{\tau^{(i)} < \infty\}} \mathcal{L}^{(i)}$$

- Reweighting by $\mathcal{L}^{(i)}$ ensures the estimator remains **unbiased**
- Variance is reduced because more paths contribute non-zero payoffs

Simulation Results — Comparison

Table: Standard Monte Carlo

Metric	Value
Analytical price	0.2134
BGK corrected price	0.1774
Estimated price	0.1740
Standard error	0.0063
95% CI	[0.1617, 0.1863]
Knocked-in paths	39.20%

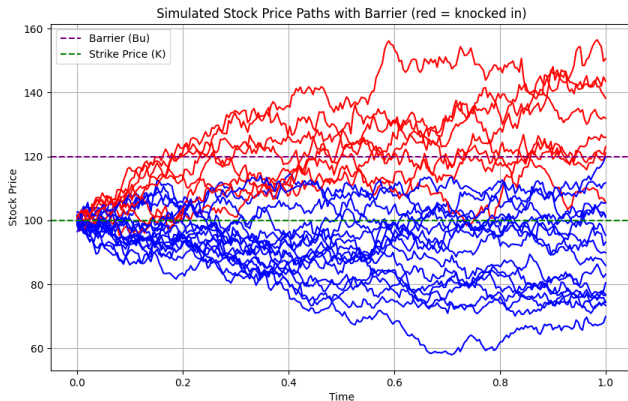
Table: Importance Sampling

Metric	Value
Analytical price	0.2134
BGK corrected price	0.1774
Estimated price	0.1788
Standard error	0.00098
95% CI	[0.1769, 0.1807]
Knocked-in paths	89.81%

Key takeaway

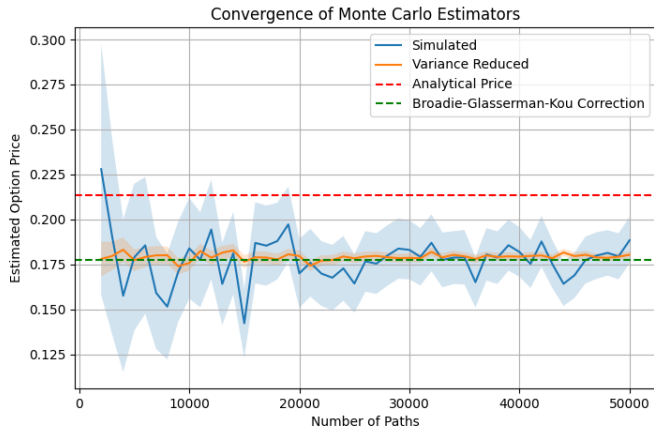
IS reduces the standard error by a factor of $\approx 6.4\times$ and more than doubles the fraction of informative (knocked-in) paths.

Simulated Stock Price Paths



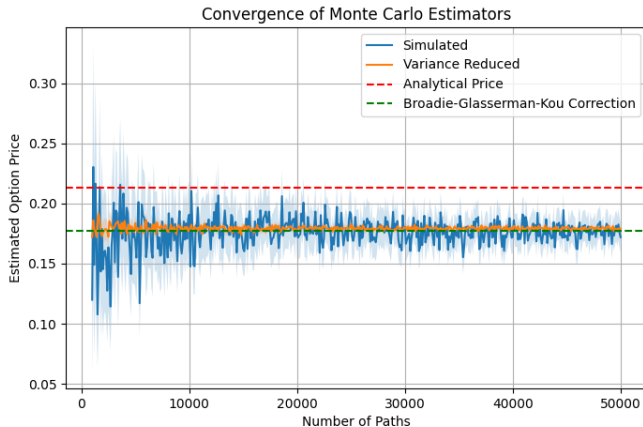
Red paths hit the upper barrier $B_u = 120$ (knocked-in); *blue paths* expire worthless. The dashed lines mark B_u (purple) and the strike K (green).

Convergence & Confidence Intervals



Convergence of both estimators to the BGK-corrected price. The IS estimator (orange) has significantly **narrower confidence bands** throughout.

Convergence — Finer Discretization



With finer path-count intervals, the superior stability of the IS estimator is even more evident — achieving comparable accuracy with far fewer simulated paths.

Conclusions

- Importance sampling is a **powerful and unbiased** variance reduction technique for rare-event pricing
- The two-phase drift shift **directly targets** the joint event that generates a non-zero payoff
- SE reduced $\approx 6\times$ vs. standard MC with the **same number of paths**
- IS paths converge to the **BGK-corrected** analytical price, confirming correctness of the discrete-monitoring adjustment
- The same framework can be applied to **all types** of barrier options

Efficiency gain

$$\frac{SE_{MC}}{SE_{IS}} \approx 6.4\times$$

Same accuracy in $\sim 40\times$ fewer paths

Thank you for your attention.